AI Expense Auditor

A smart, full-stack solution designed to automate financial auditing and receipt verification using Generative AI.

The Problem

Manual auditing of employee expense reports is a **slow and error-prone process**. It frequently leads to reimbursement delays and accidental policy violations. Organizations struggle to verify receipt authenticity and enforce spending rules at scale, often relying on manual data entry which is susceptible to human error.

The Solution

The **AI Expense Auditor** provides an end-to-end automated verification workflow:

- **Smart OCR:** Uses Google Gemini AI to "read" and extract data from uploaded receipts (Merchant, Amount, Date).
- **Policy Compliance:** Automatically flags suspicious or non-compliant expenses based on company rules.
- **Admin Control:** A dedicated dashboard for auditors to manually approve or reject flagged claims with custom feedback notes.
- **Employee History:** A persistent tracking system for employees to monitor the status of their past submissions.

Tech Stack

- **Frontend: React.js (Vite), Tailwind CSS v4**
- **Backend:** FastAPI (Python)
- **Database:** SQLite via SQLAlchemy ORM
- **AI Model: Google Gemini API**
- **UI Components: Lucide React (Icons), React Hot Toast**
- **Networking: Axios, React Router**

Future Improvements

While the current version is a fully functional MVP (Minimum Viable Product), given more time, the following enhancements would be implemented:

- **OCR Confidence Scoring:** Implementing a "Confidence Score" from the AI. If the AI is less than 80% sure about a receipt, it would automatically move to a "High Priority Review" queue for the Auditor.
- **Multi-Receipt Batch Uploads:** Allowing employees to drag and drop multiple receipts at once and process them without making the user wait.
- **Advanced Analytics Dashboard:** Adding charts for Auditors to visualize spending trends, such as "Top Spending Categories" or "Monthly Budget Burndown."
- **Email Notifications:** Integrating an SMTP server to automatically email an employee the moment their claim is "Approved" or "Rejected" by the auditor.